
WATER GARBAGE COLLECTOR – PROJECT REPORT

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1. Introduction

The story of our Water Garbage Collector project began with a simple observation that kept repeating itself across various water bodies around us. Whenever we passed by ponds, lakes, or canals—even the small ones inside residential areas—we noticed one thing in common: floating garbage. Plastic covers drifted like leaves, bottles bobbed gently in the water, and pieces of thermocol remained stuck at the edges. This waste did not dissolve, did not sink, and did not disappear on its own. It stayed there, disturbing aquatic life, ruining the aesthetics, and creating health hazards.

As students who observe problems naturally and try to think of solutions, we started imagining what it would take to clean such waste efficiently. We observed manual cleaning efforts where workers used long sticks or nets to pull waste out of the water. Watching them struggle made us realize that the process was slow, unsafe, and physically tiring. Workers had to lean dangerously close to polluted water, risking infections or injuries. And yet, the amount of waste collected compared to the effort put in was minimal.

This made us question: “What if there was a simple floating system—easy to use, low-cost, and mechanically effective—that could reduce the strain on workers?” That thought sparked the foundation of this project. Instead of designing a fully robotic or automated system that would be expensive and hard to maintain, especially in rural regions, we wanted something functional, practical, and realistic.

This is how our Water Garbage Collector idea came to life—a manually operable, stable floating device equipped with a mechanical conveyor mechanism that collects floating waste and directs it into a storage bin. The goal was to make it feel natural, almost like extending a helpful tool into the hands of existing cleaning workers, but without requiring them to touch polluted water or exert heavy physical effort.

From that point forward, the project grew step-by-step, evolving into a working prototype built with careful selection of materials, appropriate components, and thoughtfully designed mechanical structures. The following sections narrate that journey—from concept to completion— as an understandable, relatable, and engineering-strong story.

2. The Evolution Of The Design

Design evolution is where the real story of our system begins. Once we knew what we wanted to build, the next challenge was deciding how it should look and function. A floating system may sound simple, but factors such as balance, buoyancy, load distribution, motor placement, and water resistance make the design complex.

Initially, we considered making the system in the shape of a small boat. However, we soon realized that this design was unbalanced and not suitable for holding trash effectively. To build our prototype quickly and test the concept, we opted to use a plastic box as the floating base.

Ideally, a catamaran-style (dual-hull) design would provide natural stability, with two parallel supports distributing the center of gravity evenly and keeping the system upright even when the bin is full. While this would be the best long-term solution, we did not implement it in our current prototype.

For the prototype, we focused on practical testing: we experimented with conveyor angles, ensuring that waste would glide smoothly into the collection bin without slipping or getting stuck. This hands-on iteration helped us refine component placement, floating base dimensions, and wiring layout, turning sketches into a functional working model.

3. Components Used

A system is only as effective as the components used to build it. Every component in our Water Garbage Collector was chosen not only because it performs the required function, but also because it fits our core design philosophy: simplicity, affordability, and durability.

1. **Arduino Uno** – Acts as the brain of the system. It sends PWM signals to control the motor speeds and directions.

Arduino Model	UNO R3
Microcontroller	ATmega328P
Operating Voltage	5V
Supply Voltage (recommended)	7-12V
Maximum Supply Voltage	20V
Digital I/O Pins	14 (of which 6 are PWM pins)
Analog Input Pins	6
DC Current per I/O Pin	40mA
DC Current for 3.3V Pin	50mA
Flash Memory	32 KB of which 0.5 KB is used by the Bootloader
SRAM	2KB
EEPROM	1KB
Clock Speed	16MHz

2. **Motor Drivers (Two L298N Modules)** – One driver controls the two propulsion motors responsible for forward, backward, and turning motions. The second driver controls the conveyor motor.

Driver Model	L298N 2A
Driver Chip	Double H-Bridge L298N

Motor Supply Voltage (Maximum)	46V
Motor Supply Current (Maximum)	2A
Logic Voltage	5V
Driver Voltage	5-35V
Driver Current	2A
Logic Current	0-36mA
Maximum Power (in Watts)	25W
Current Sense for Each Motor	Yes
Heat Sink	Yes
Power ON LED Indicator	Yes
Weight (in grams)	25g
Dimensions (millimeters)	44x44x28 (LxBxH)

3. **Four DC Motors (with shafts)** – Two motors are responsible for propulsion, and two motors drive the conveyor system.

RPM	100
Voltage	3-12V
Output Torque	3.5 Kg-Cm
Weight	200g

4. **Floating Base** – A waterproof plastic base that provides buoyancy and stability.

Material	Plastic
Dimensions (in Cm)	23x16x5 (LxBxH)

5. **Conveyor Belt Mechanism** – The mechanism responsible for lifting waste from the water surface into the bin.

Material	Plastic
Length	15 Cm
Angle	35 Degree

6. **Storage Bin** – A lightweight removable bin that collects waste.

Material	Plastic
Dimensions (in Cm)	6x15.5x5 (LxBxH)

7. **Battery** – A 12V battery pack with AA batteries providing power to motor drivers and a 9V Battery for Arduino.

Battery Holder (two in Series for 12V)	
Compatible Battery Size	AA Battery
Material	Plastic
Internal Connections	Series
Cable Length (in Cm)	15 Cm
Dimensions	62 x 58 x 15mm
Weight	15 grams
Capacity	4 cells

9V Battery	
Discharge Time	27 ohm, 9 Hours
Nominal Voltage (V)	9
Jacket Material	Metal
Discharge Terminal Type	Press Stud Terminal
Dimensions (mm) L x W x H	4.5 x 2.2 x 1.5
Weight (gm)	178

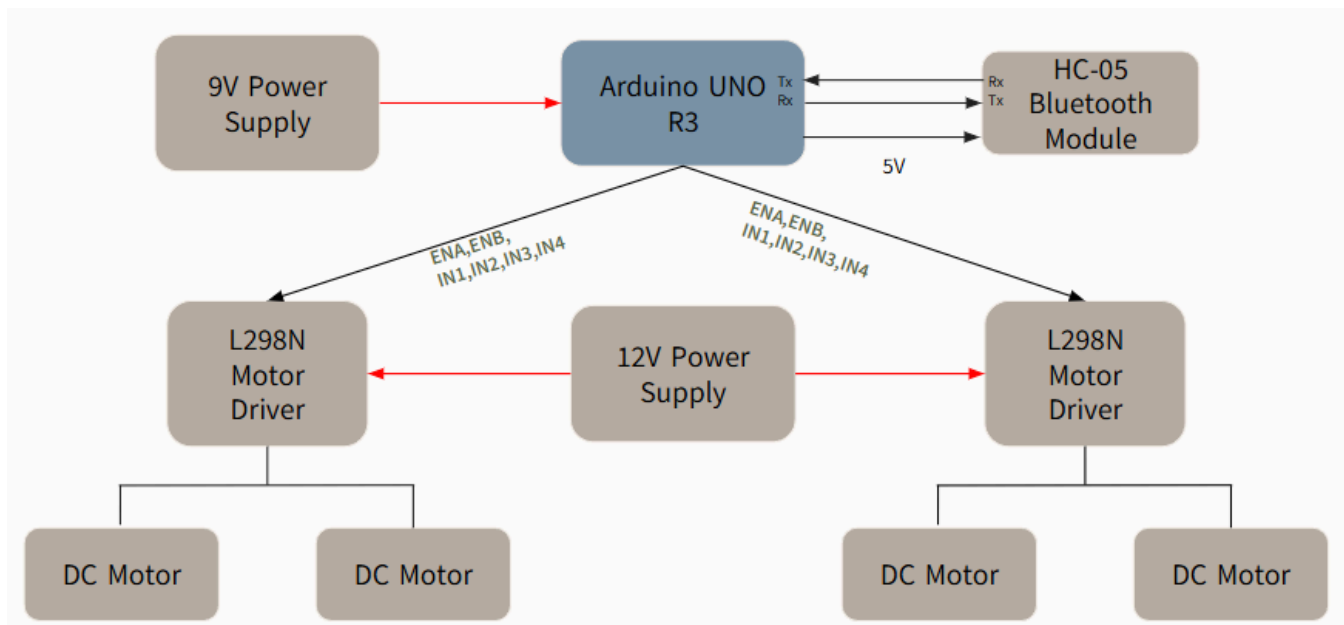
8. **HC-05 Bluetooth Module** - Used for communication between a smartphone and the robot.

Serial Communications	9600-115200bps
Frequency	2.4~2.524GHz

Power Supply	5V DC
Operating Current	30mA
Range	<100m
Dimensions	30mm x 14mm x 2.2mm
Weight	5 grams

Each of these components works in coordination to create a system that is robust yet simple enough to be maintained even in rural or low-budget environments.

4. Block Diagram of Water Garbage Collector



The overall system is built around the Arduino Uno R3, which functions as the main control unit. It receives user commands through the HC-05 Bluetooth module, which communicates with the Arduino via serial Tx-Rx pins. The Bluetooth module is powered directly from the Arduino's 5V output, ensuring stable communication.

Two separate power supplies are used in the system:

- A 9V supply powers the Arduino Uno R3 and provides the logic-level power required for processing and communication.

- A 12V supply powers both of the L298N motor driver modules, since the DC motors require a higher operating voltage and current than the Arduino can provide.

The Arduino sends control signals to each L298N motor driver using the pins ENA, ENB, IN1, IN2, IN3, and IN4. These signals determine the speed and direction of the connected motors.

Each L298N motor driver controls two DC motors, giving a total of four motors in the system. One motor driver is dedicated to the propulsion motors, enabling forward, backward, and turning motion. The second motor driver powers the conveyor motors responsible for lifting waste into the bin.

Overall, the block diagram shows a clear flow of power and control:

- Power supplies feed the Arduino and motor drivers.
- The Arduino processes Bluetooth commands.
- The Arduino outputs motor control signals to the drivers.
- The drivers regulate the motors that move and operate the garbage-collection mechanism.

This structure ensures that the system is electrically stable, modular, and efficient in handling both control logic and high-power motor operation.

5. Experimental Setup

The experimental setup was intended to simulate real-life conditions. However, we did not physically test the system in water. Instead, we focused on building a fully functional circuit and prototype model.

Through careful assembly and testing of the components, we established that the model operates correctly. The propulsion and conveyor motors respond as intended, and the Bluetooth connectivity works reliably without any issues. While actual water testing was not conducted, the system's functionality and control have been successfully verified in a controlled setup.

6. Work Done

The work done throughout the project can be described as a series of meaningful steps:

1. **Problem Identification** – Understanding the inefficiency of manual waste collection and recognizing the need for a support system.
2. **Component Selection** – Choosing cost-effective, durable, and functional components.
3. **Structural Fabrication** – Building the floating base, aligning shafts, waterproofing joints.
4. **Mechanical Assembly** – Installing the conveyor belt, adjusting angles, ensuring smooth rotation.
5. **Electrical Wiring** – Connecting motor drivers, Arduino pins, and securing insulated wiring.
6. **Programming using Arduino IDE** – Writing code to control propulsion and conveyor operation.

Each stage taught us something new and allowed us to refine the system for better performance.

7. Results And Analysis

Although the Water Garbage Collector prototype was not tested in actual water, the system performed well during controlled dry testing. The propulsion motors, when powered, responded correctly to Bluetooth commands, demonstrating smooth forward, backward, and turning actions. The conveyor mechanism rotated reliably, confirming that the mechanical assembly and motor drivers were functioning as intended.

The dual-motor-driver configuration ensured stable power delivery to all motors, and the Bluetooth module connected without interruptions throughout operation. These successful results indicate that the prototype's electrical, mechanical, and control systems are working as designed.

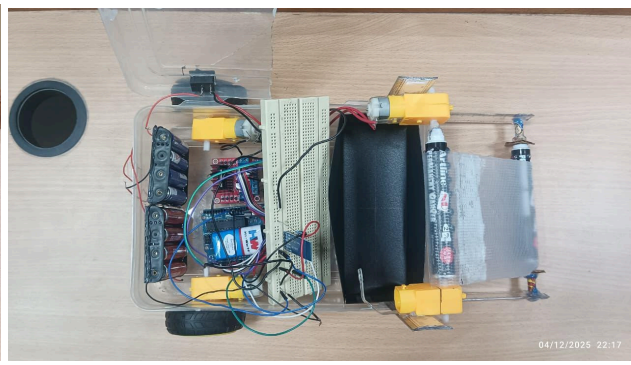
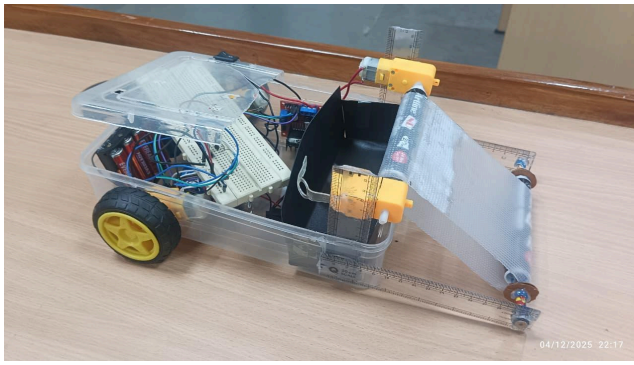
Key Observations:

- The motors responded accurately to directional commands during dry testing.
- The conveyor assembly rotated smoothly with consistent speed.
- Bluetooth pairing was stable, with no signal drop or delay.
- The structural model supported the components securely without imbalance.

Limitations:

- The prototype was not tested in water, so buoyancy and load behaviour remain unverified.
- The waste-lifting efficiency of the conveyor in real water conditions could not be evaluated.
- The current model still requires manual guidance via Bluetooth.

Despite these limitations, the completed prototype successfully demonstrates the core concept. With proper waterproofing, material upgrades, and future water testing, the system can be further refined into a field-ready, practical solution for surface waste collection—especially in low-budget or rural environments.



8. Conclusion

In conclusion, the Water Garbage Collector project successfully demonstrated the feasibility of a simple, low-cost waste-collection system through a fully functional prototype. Although the system was not tested in water, the completed circuit, motor control logic, conveyor mechanism, and Bluetooth operation worked reliably during dry testing. This confirms that the core idea—collecting floating waste using a motorised conveyor on a stable floating platform—is practical and achievable.

The prototype highlights how basic engineering principles and thoughtful component selection can contribute to solving real environmental challenges. While it is not yet a deployable field model, it establishes a strong foundation for further development. Future improvements such as waterproofing, full water-based testing, solar integration, remote control, and enhanced propulsion can transform this prototype into an effective real-world cleaning system.

Overall, the project meets its objective of proving the concept and creating a functioning model that can be developed into a complete, operational solution with further refinement.

9. Complete Code

The program for the system was developed using the **Arduino Integrated Development Environment (IDE)**. The IDE was used to write the source code, manage project configurations, and compile the programme into machine-level instructions for the Arduino Uno R3. Once compiled, the code was uploaded to the microcontroller using a **USB Type-A to USB Type-B cable**, with the Type-A end connected to the computer and the Type-B end connected to the Arduino board.

```
#include <SoftwareSerial.h>
```

```
int motor1pin1 = 2;
```

```
int motor1pin2 = 3;
```

```
int motor2pin1 = 4;
```

```
int motor2pin2 = 5;
```

```
int enA = 9;
```

```
int enB = 10;
```

```
int motor3pin1 = 1;
```

```
int motor3pin2 = 0;
```

```
int enC = 6;
```

```
int motor4pin1 = 7;
```

```
int motor4pin2 = 8;
```

```
int enD = 11;
```

```
SoftwareSerial bt(12, 13);
```

```
void drive(int sA, int sB, int p11, int p12, int p21, int p22) {
```

```
    analogWrite(enA, sA);
```

```
    analogWrite(enB, sB);
```

```
    digitalWrite(motor1pin1, p11);
```

```
    digitalWrite(motor1pin2, p12);
```

```
    digitalWrite(motor2pin1, p21);
```

```
    digitalWrite(motor2pin2, p22);
```

```
}
```

```
void stopMotors() {
```

```
    drive(0, 0, 0, 0, 0, 0);
```

```
}
```

```
void conveyer(int state) {
```

```
    int sp = state * 120;
```

```
analogWrite(enC, sp);  
analogWrite(enD, sp);  
digitalWrite(motor3pin1, state);  
digitalWrite(motor3pin2, !state);  
digitalWrite(motor4pin1, state);  
digitalWrite(motor4pin2, !state);  
}
```

```
void setup() {  
  pinMode(motor1pin1, OUTPUT);  
  pinMode(motor1pin2, OUTPUT);  
  pinMode(motor2pin1, OUTPUT);  
  pinMode(motor2pin2, OUTPUT);  
  pinMode(enA, OUTPUT);  
  pinMode(enB, OUTPUT);
```

```
  pinMode(motor3pin1, OUTPUT);  
  pinMode(motor3pin2, OUTPUT);  
  pinMode(enC, OUTPUT);
```

```
  pinMode(motor4pin1, OUTPUT);  
  pinMode(motor4pin2, OUTPUT);  
  pinMode(enD, OUTPUT);
```

```
  Serial.begin(9600);  
  bt.begin(9600);  
}
```

```
void loop() {  
  if (bt.available()) {
```

```

char c = bt.read();

if (c == 'w') drive(120, 120, 1, 0, 1, 0);
else if (c == 's') drive(120, 120, 0, 1, 0, 1);
else if (c == 'a') drive(90, 120, 0, 1, 1, 0);
else if (c == 'd') drive(120, 90, 1, 0, 0, 1);
else if (c == 'x') stopMotors();
else if (c == 'u') conveyer(1);
else if (c == 'v') conveyer(0);
}
}

```

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